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Development of Information System Student Period Completion Mobile-Based

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ABSTRACT
This research aims to develop a mobile-based student study period completion information system in JPTE. The system developed uses the fountain development model approach with 5 stages, namely analysis, subsystem requirements, design, implementation, and testing. The purpose of choosing the fountain model is to ensure flexibility in development and adaptation to user needs. Black Box testing is carried out to ensure that each main feature, such as login, thesis/final project upload, journal upload notification, input and display of graduation dates, as well as management of student and KAPRODI data, functions according to the specifications set. The test results show that all 100% of the features successfully meet the expected functional criteria, with a system that is able to provide easy access, flexibility, and responsiveness. Overall, the Black Box test results show that the system is ready to be used by users and meets all the functional requirements that have been set. This mobile-based information system is expected to improve the efficiency of academic administration management and support the smooth process of completing the student study period.

Keywords: Information System; Fountain; Mobile

1. Introduction

Along with the times, the use of information systems has become increasingly integral in various fields of life [1]. In this digital era, almost all sectors, including business, government, and education, have shifted from manual methods to more efficient and effective information systems [2]. Information systems not only increase the speed and accuracy of data processing, but also allow real-time access to information, which is very important in making quick and precise decisions [3][4]. Realizing the importance of information systems in improving the efficiency and effectiveness of data management, almost all universities in Indonesia have adopted and integrated information systems in data management and academic administration. The efficiency of academic data management is very important for the success of the education process. In a dynamic academic environment, efficient and accurate data management is the main key to support various academic and administrative activities [5].

The Department of Electrical Engineering Education (JPTE) UNM has a website-based information system. Based on the observation, it is known that JPTE has implemented an information system to manage various aspects of administration, such as lecturer data, student data, correspondence management, mentoring, tracer studies, and student seminars. The implementation of this information system has provided significant benefits in expediting the administrative process in the department. However, there are still some shortcomings, including the absence of features that allow users to view student journals, monitor the study period, and check the GPA of each student comprehensively. In addition, the entire information system is currently web-based, so it needs to be further developed into a mobile version.

Mobile-based information systems have several advantages over website-based information systems. First, mobile-based information systems allow more flexible access with smartphones [6]. This allows users to get information more easily without being limited by desktop devices. Another advantage of mobile-based information systems is their ability to provide a better user experience through a responsive user interface [7]. Thus, mobile-based information systems offer advantages in terms of accessibility, user experience, and ease of use compared to web-based information systems [8]. The use of mobile-based information systems can provide greater flexibility for users and increase efficiency in various contexts of use. Based on the above problems, the researcher wants to develop a mobile-based information system for completing the study period at JPTE.

2. Materials and Methods

This type of research is Research and development (R&D) with the fountain development model. The *fountain* development model is a logical improvement of the waterfall model [9]. Although the steps and procedures are still the same, in the process this *fountain* model is more flexible and responsive to changing needs because it allows developers to prioritize

or and skip some steps, except for certain stages, such as design before coding, cannot be skipped to avoid overlaps that can occur in the development cycle.

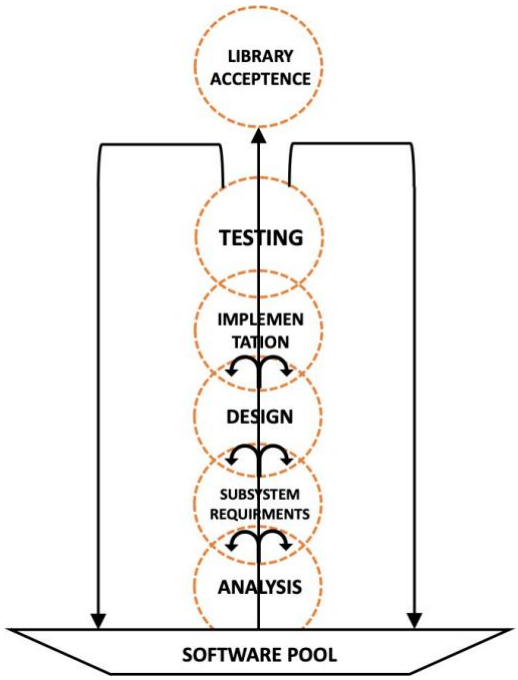


Figure 1: Stages of Development of Student Study Completion Management Information System

There are several stages in the *fountain* model, but in this research, the stages are simplified according to the needs of the research, which are as follows:

a. Analysis: Analyzing the current system by gathering information from users to understand their needs and expectations of the mobile version of the system to be developed.
b. Subsystem Requirements: Identified new requirements for the mobile version by designing the functional and nonfunctional requirements of the system. The functional requirements of this system are described in more detail using the following use case diagram and activity diagram. Use Case Diagram is a technique for recording the functional requirements of a system, describing the expected functionality of a system. Use case diagrams emphasize "what" is done by the system, and not "how". A use case represents an interaction between an actor and the system [10]. This use case can be said to be a description of the functionality of the system from the user's point of view, then the activity diagram will be drawn. Activity diagrams serve to describe the various activity flows in the system being designed, how each flow begins, decisions that may occur, and how they end. Activity diagrams can also describe parallel processes that may occur in multiple executions [11].
c. Design: Design the system interface for the mobile version.
d. Implementation: Implementing the design into program code. The application backend will be developed using the Java programming language with the google firebase database. Application development will use various tools such as android studio, Git for version control, and npm or pip for package management.
e. Testing: Conducting tests to ensure that the system meets the requirements specifications and works correctly. Testing is done using the black box method.

3. Results

a. Analysis

Based on the results of the researchers' observations, information was obtained that the Electrical Engineering Department has implemented the use of information systems to manage administration including data on lecturers, students, correspondence management, mentoring, tracer studies and student seminars. However, there is no feature to view the overall student journal, study period and GPA of each student. Furthermore, all information systems in the Electrical Engineering Department are website-based and need to be developed again for the mobile version.

b. Subsystem Requirements

Based on the results of the analysis conducted, additional features will be made in the developed mobile application. Where this application allows students to upload final assignments and journals, input their graduation dates, and view their academic progress. Overall, the process of managing this application is classified into 3.

Table 1. User Description

Username	Definition
Admin	▪ Can insert, delete and edit student and KAPRODI data Can see the student's study period. ▪ Can insert, delete and edit user accounts
Head of Study Program (KAPRODI)	▪ Can insert, delete and edit student data ▪ Can update the journal uploaded by students Can see the student's study period.
Student	▪ Can edit student data. ▪ Can upload journals/final assignments. ▪ Can view journal approval notifications that are acceded by KAPRODI

1) System Functional Requirements

System functional requirements are used to identify system functional requirements that must be met based on user needs. In general, the functional requirements of this system can be seen in the following table.

Table 2. Functional Requirements of the system

No.	Functional Requirements	Explanation
1	Login & Forgot password	Users must be able to log into the application and if they forget their password they can access forgot password.
2	Upload thesis/final project	Students can upload their thesis or final project and journal into the system.
3	Notification after journal upload	Students receive a notification after their journal is approved or rejected by KAPRODI.
4	Manage study period data	Students, KAPRODI and admin can manage a summary of the progress of student study periods such as inputting graduation dates, editing, adding and deleting in the form of a dashboard.
5	Approve and reject journals	KAPRODI can approve or reject journals uploaded by students.
6	Manage student data	KAPRODI, admin and students can manage student data such as update, edit, and delete student data.
7	Manage KAPRODI data	Admin can manage KAPRODI data such as update, edit, and delete KAPRODI data.
8	Manage user accounts	Admins must be able to manage user accounts such as update, edit, and delete user accounts.

As has been explained that in this system, there are 3 users who act as Actors, namely Admin, KAPRODI and Students. While the use cases or things that can be done by actors in this system consist of 8, namely Managing User Accounts, Managing Student Data, Managing KAPRODI Data, Accessing Journal Lists, Managing Study Period Data, Upload Skirpsi and Journal, and Viewing Notofication. In more detail, the relationship between Actor and Use Case can be seen in Figure 2.

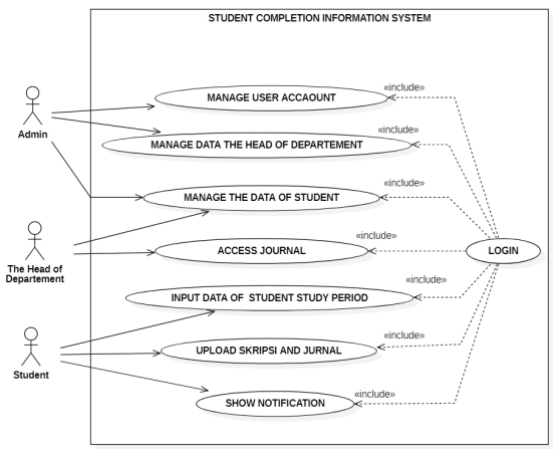


Figure 2. Use case diagram

a) Admin Use Case

In this Admin use case, the admin can login with his account. If the login is successful, the admin will enter the main page, but if the login fails, the admin will be directed back to the login page and access forgot password. After successfully logging in, the admin can view and edit student study period data which includes GPA, SKS, graduation date, and student study period. Admins can also manage student and KAPRODI data, including adding, deleting, and editing student and KAPRODI data. In addition, admins can manage managing user accounts where admins can add, delete and edit.

Table 3. Description of the admin use case

Action by Admin	Response from system
Login and forgot password	The system analyzes whether it is true or false, if successful then the admin can log in, however, if there is an error it will be returned to the login page and the admin can access the forgot password feature.
Manage student study period data	The system will direct the admin to be able to see the student's GPA, SKS, study period and graduation date as well as edit, add and delete student data.
Manage KAPRODI data	Admin can add, edit and delete KAPRODI data on the system.

b) KAPRODI Use Case

In this KAPRODI use case, KAPRODI can login with his account. If the login is successful, KAPRODI will enter the main page, but if the login fails, KAPRODI will be directed back to the login page and access forgotten access. After successfully logging in, KAPRODI can view the student study period which includes GPA, SKS, graduation date, and student study period. KAPRODI can also manage student data, including adding, deleting, and editing student data. In addition, KAPRODI can manage student journal results that have been uploaded, with the ability to view and approve or reject the journal.

Table 4. Description of the KAPRODI usecase

Action by KAPRODI	Response from system
Login and forgot password	The system analyzes whether it is true or false, if it is successful then KAPRODI can enter, however, if there is an error it will be returned to the login page and the admin can access forgotten password
Manage student study period data	The system will direct KAPRODI to be able to see GPA, SKS, study period and student graduation date as well as edit, add and delete student data.
Manage Student Data	KAPRODI can add, edit and delete student data.
Manage Journal Results	KAPRODI can view and approve or reject journals that students have uploaded to the system.

c) Student Use Case

In this student use case, students can login with their account. If the login is successful, students will enter the main page, but if the login fails, students will be directed back to the login page. After successfully logging in, students can see the student study period which includes GPA, SKS, graduation date, and student study period. Students can also manage student data, including adding, deleting, and editing student data and inputting graduation dates. In addition, students can see journal notifications that have been approved by KAPRODI.

Table 5. Description of the Student Usecase

Action by Student	Response from system
Login and forgot password	The system analyzes whether it is true or false, if it is successful then the student can enter but, if there is an error, it will be returned to the login page and the admin can access the forgotten password.
Manage student study period data	The system will direct students to be able to see GPA, SKS, study period and student graduation date as well as edit, student data.
Uploading Thesis and Journal	Students are directed to the place of uploading journals or theses by the system
View Notification of Journal and Thesis Uploading	The system will send a notification to the student if KAPRODI has approved or rejected the uploaded journal.

d) Activity Diagram

To understand the process in this system as a whole, 8 activity diagrams were created, as follows:

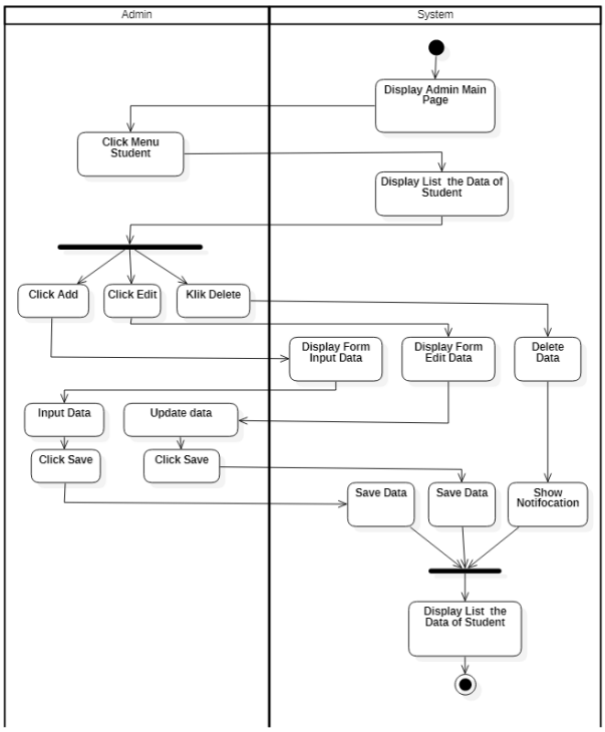


Figure 3. Activity Digram Manage Student Data

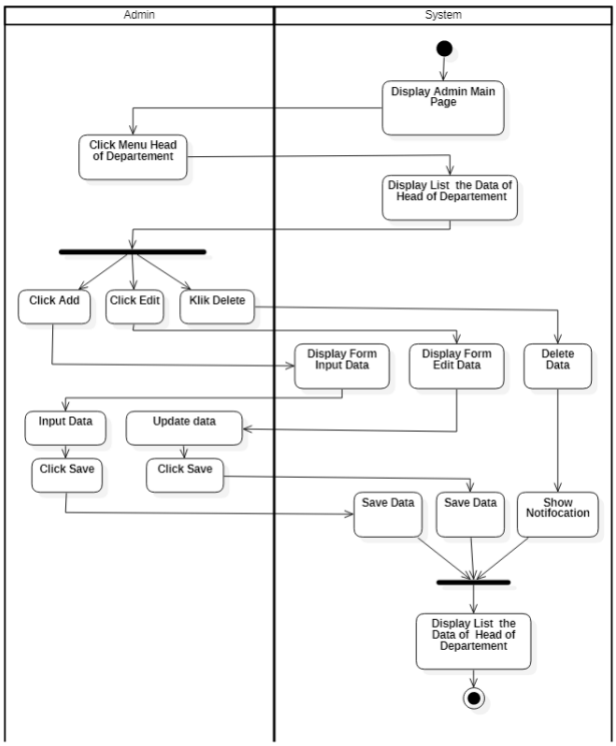


Figure 4. Activity Digram Manage KAPRODI Data

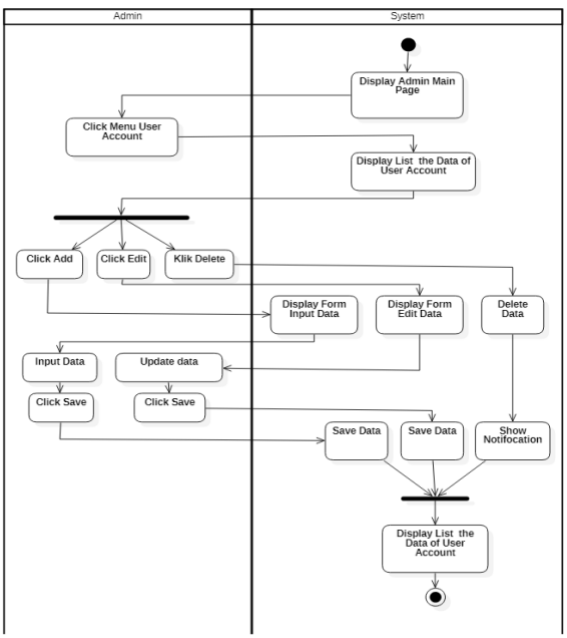


Figure 5. Activity Digram Manage User Account Dat.

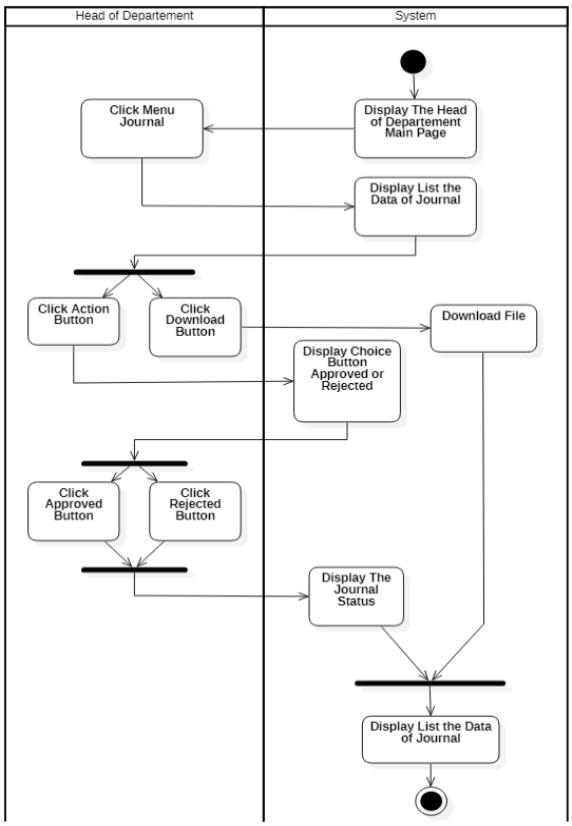


Figure 6. Activity Diagram of Journal Validation

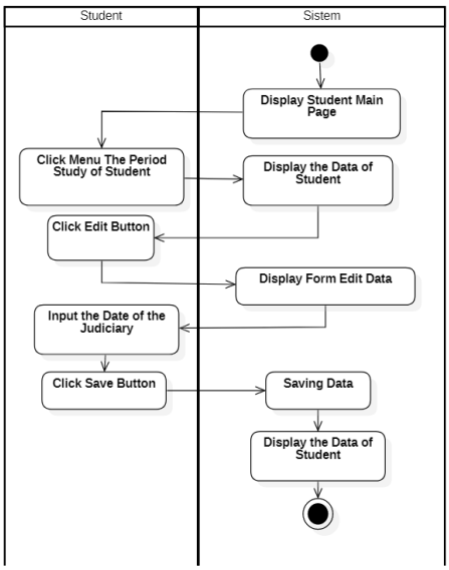


Figure 7. Activity Diagram of Study Period Input.

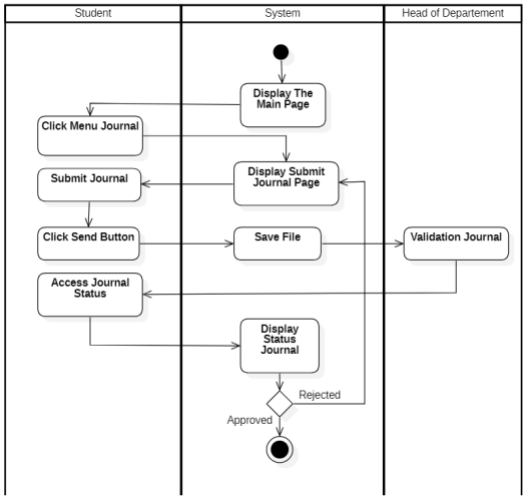


Figure 8. Activity Diagram of Thesis and Journal Upload

2) Non-Functional System Requirements

Nonfunctional requirements in this system describe the quality and technical limitations of the system, such as availability, reliability, ergonomy, portability, memory, response time, safety, and security of the system to be created.

Table 6. Non-Functional System Requirments

Num.	Parameters	Needs
1	Availability	The app must be available 24 hours a day, 7 days a week without interruption.
2	Reliability	The application must have a high level of reliability, with a minimal failure rate.
3	Ergonomy	The user interface should be designed for user convenience and easy to use.
4	Portability	The application should be able to be easily installed and operated on various mobile devices with the Android operating system.
5	Memory	Memory usage should be optimized for optimal performance on mobile devices
6	Response time	The application must respond in a fast time, for example, displaying a page within 3 seconds of a user request.
7	Security	User data must be secured with strong encryption and a secure authentication system
8	Others 1	Communication language | All user interactions within the app must be in Indonesian.

c. Design

The following is the user interface of the developed application:



Figure 9. User Interface Design

d. Implementation

The application that has been developed can run on the Andorid operating system. Here are the results of the application development:



Figure 10. Login Page



Figure 11. Admin Main Page



Figure 12. KAPRODI Main Page

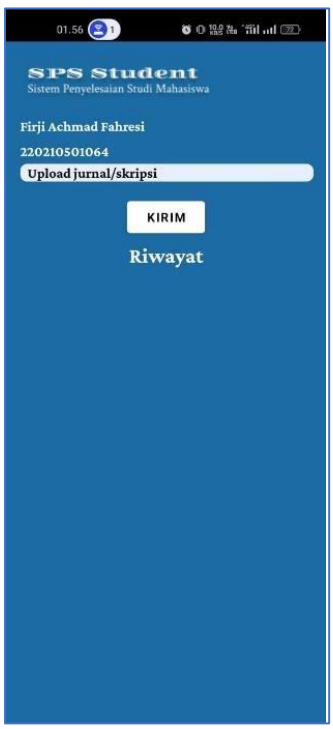


Figure 13. Journal Upload Page



Figure 14: Journal Validation Page

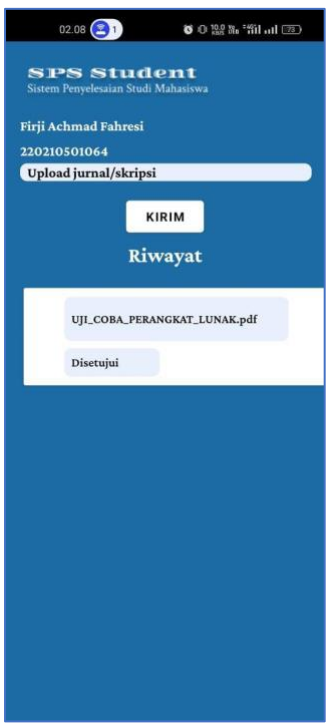


Figure 15: Journal Status Page

e. Testing

The app was tested using Android and iOS smartphones. Android devices require at least one smartphone of the latest generation, such as Samsung Galaxy S21 or Google Pixel 6, and one of the older generations, such as Samsung Galaxy S9 or Google Pixel 3, with an OS version of at least Android 8.0 (Oreo) or later. For iOS devices, at least one iPhone of the latest generation, such as iPhone 13 or iPhone 12, and one of the older generations, such as iPhone 8 or iPhone SE, with an OS version of at least iOS 12 or later are required. The following are the results of functional testing of the application using the black box method by 3 software testers (T1, T2 and T3).

Table 7. Black Box Testing Results

NO	Scenario Testing	Test Case	Results. expected	Test Results		
				T1	T2	T3
1	Login	User enters username and password correctly	The user successfully logs into the app and is directed to the main dashboard.	✓	✓	✓
		User enters incorrect username and password	Login failed and an error message appears.	✓	✓	✓
2	Upload thesis/final project	Students upload journal files	File uploaded successfully	✓	✓	✓
3	Journal notification	Journal successfully uploaded	Notification received by the user with the message "Journal successfully uploaded"	✓	✓	✓
4	Input and view judicium dates	User inputs the date of graduation	The judicium date is successfully saved and displayed correctly.	✓	✓	✓
5	Manage study period summary	User inputs study period summary	The study period summary was successfully updated and displayed correctly.	✓	✓	✓
6	Approve/reject the journal	KAPRODI selects the agree button	Journal approved updated status.	✓	✓	✓
		KAPRODI selects the disagree button	The journal is rejected, and the status is updated.	✓	✓	✓
7	Manage student data	Add student data	Student data was added successfully.	✓	✓	✓
		Editing student data	Student data was successfully edited.	✓	✓	✓
		Delete student data	Student data is successfully deleted according to the action taken.	✓	✓	✓
8	Manage KAPRODI data	Adding KAPRODI Data	KAPRODI data added successfully.	✓	✓	✓
		Editing KAPRODI Data	KAPRODI data was successfully edited.	✓	✓	✓
		Delete KAPRODI Data	KAPRODI data is successfully deleted according to the action taken.	✓	✓	✓
9	Manage user accounts	Add a new user	User account added successfully.	✓	✓	✓
		Editing user data	User account edited successfully.	✓	✓	✓
		Delete user data	User account successfully deleted.	✓	✓	✓

4. Discussion

In this research, the Fountain Model is used as an approach to develop a mobile-based student study period completion information system. The fountain development model is more flexible and adaptive to changing needs and user feedback. According to [12] that one of the advantages of the fountain model is enhanced flexibility in changing requirements. In this study, changes or additions to the journal status notification feature requested by users can be accommodated immediately without disrupting the overall development flow, thanks to the iterative nature of this model. This information system integrates various features that were previously unavailable in a web-based system, such as features to view student journals as a whole, monitor the study period, and check the GPA of each student comprehensively. One of the advantages of this information system is the user interface that is friendly and easy to use. In line with research [13] which says that mobile-based information systems facilitate access to academic information that is more efficient than access via the web which requires typing the site address and a user interface that is not optimized. Mobile-based information system users provide advantages This integration provides a more holistic experience for users, allowing them to access all the information needed using a smartphone.

Black box testing can find errors or mismatches between input and output. According to [14] The main purpose of testing is to design tests that systematically uncover types of errors that have not been detected before. Testing is carried out by 3 validators without seeing the program source code and testing based on scenarios and test cases that have been made by researchers. According to [15] black box testing has advantages in its ability to find errors and problems without requiring an understanding of the internal workings of the software. The test results show that all 100% of the system features function according to the expected specifications. According to [16] Black box testing is a technique used to confirm that an application's functionality satisfies requirements by examining how the application executes based on the input it receives. Overall, the results of Black Box testing show that this system is ready to be used by users and meets all predefined functional requirements.

5. Conclusions

This research successfully developed a mobile-based student study period completion information system using the Fountain Model as a development approach. The fountain model is more flexible and adaptive to changing needs and feedback from users. Black Box testing shows that all system features function according to the expected specifications with a 100% success rate, so this system is feasible and ready to use.

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